Bryce Cook

brycecook2018@gmail.com | (931)335-1485 | Portfolio | github.com/bcook26

Skills

Programming Languages: Python, SQL, Bash/Shell, C++
Data Engineering: Apache Airflow, Apache Spark, PySpark, ETL/ELT, Batch Processing, Data Warehousing, Data Modeling
Databases: PostgreSQL, Oracle, MySQL, SQL Server, Database Design, Dimensional Modeling
Cloud & DevOps: AWS, Docker, Git, GitHub, AWX
Libraries & Frameworks: Pandas, SQLAlchemy, FastAPI, Streamlit

Experience

Data Engineer, Knoxville Utilities Board | Knoxville, TN May 2023 – Present

- Engineered end-to-end ETL pipelines in Python to support annual gas demand forecasting, integrating operational and weather datasets into a self-service Streamlit application used to guide infrastructure investment decisions exceeding $10M.
- Designed automated model lifecycle workflows, including model drift monitoring and retraining processes, improving forecast accuracy and reducing manual maintenance.
- Developed batch ETL workflows to ingest and warehouse fiber bandwidth telemetry from API endpoints, implementing automated scheduling, logging, and historical backfill capabilities to support machine learning initiatives.
- Built cross-functional data products using Python and SQL to consolidate meter data from multiple enterprise systems, providing real-time operational visibility for field technicians, analysts, and support teams.
- Redesigned customer billing anomaly detection workflows by optimizing SQL extraction logic and migrating spreadsheet-based transformations into Python ETL processes with automated reporting and audit logging.
- Modernized legacy reporting infrastructure by migrating multiple MS Access-based workflows into Apache Airflow-managed ETL pipelines, improving reliability and reducing manual effort across utility operations reporting.
- Automated monthly anomaly reporting and stakeholder notifications through AWX orchestration, eliminating manual report preparation and distribution processes.
- Established GitHub development standards across the organization, creating repository templates, commit conventions, and branching strategies to improve code quality and collaboration.
- Led enterprise workflow orchestration modernization by migrating scheduled jobs from Windows Task Scheduler to AWX and subsequently Apache Airflow, enabling centralized monitoring, version control integration, and secrets management.

Projects

Full Stack Golf ETL

- Developed a Python-based ETL pipeline to ingest and transform professional golf tournament statistics from public APIs.
- Designed PostgreSQL data models and automated data refresh workflows using Apache Airflow.
- Built a FastAPI backend and interactive frontend to visualize player performance metrics and historical trends.

Student Consultant (Team Lead) | August 2021 - May 2022 | Clinton, TN Duraline Manufacturing

- Redesigned production scheduling with a Python-based heuristic model, improving efficiency around the process bottleneck by approximately 6%
- Authored user documentation and training manuals, resulting in $125,000/year in recovered value-added time
- Led weekly presentations of project progress and results to senior staff and internal project stakeholders

Education

University of Tennessee – B.S. in Industrial & Systems Engineering June 2022 | Major GPA: 3.6